Chapter 1. Characterization of Distributed Systems

DISTRIBUTED SYSTEMS DEVELOPMENT

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Distributed Systems: Concepts and Design

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Objectives

- To be aware of the characteristics of concurrency, independent failure of components and lack of a global clock, which necessarily arise in a distributed system consisting of components that coordinate their actions only by passing messages.
- To place distributed systems in a realistic context through examples: the Internet, an intranet and mobile computing.

Objectives

- To motivate the benefits of resource sharing and to introduce the Web as an example.
- To gain a good understanding of the challenges related to heterogeneity, openness, security, scalability, failure handling, concurrency and transparency as they apply to distributed systems.

1.1 Introduction
Definition Of Distributed Systems

- What is a distributed system?
  - A distributed system is a collection of independent computers that appear to the users of the system as a single system.
  - As one in which hardware or software components located at networked computers communicate and coordinate their actions only by passing messages.

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1.1 Introduction
Applications of Distributed Systems

Finance and commerce	eCommerce e.g. Amazon and eBay, PayPal, online banking and trading
The information society	Web information and search engines, ebooks, Wikipedia; social networking: Facebook and MySpace.
Creative industries and entertainment	online gaming, music and film in the home, user-generated content, e.g. YouTube, Flickr
Healthcare	health informatics, on online patient records, monitoring patients
Education	e-learning, virtual learning environments; distance learning
Transport and logistics	GPS in route finding systems, map services: Google Maps, Google Earth
Science	The Grid as an enabling technology for collaboration between scientists
Environmental management	sensor technology to monitor earthquakes, floods or tsunamis

1.1 Introduction
Advantages of Distributed Systems

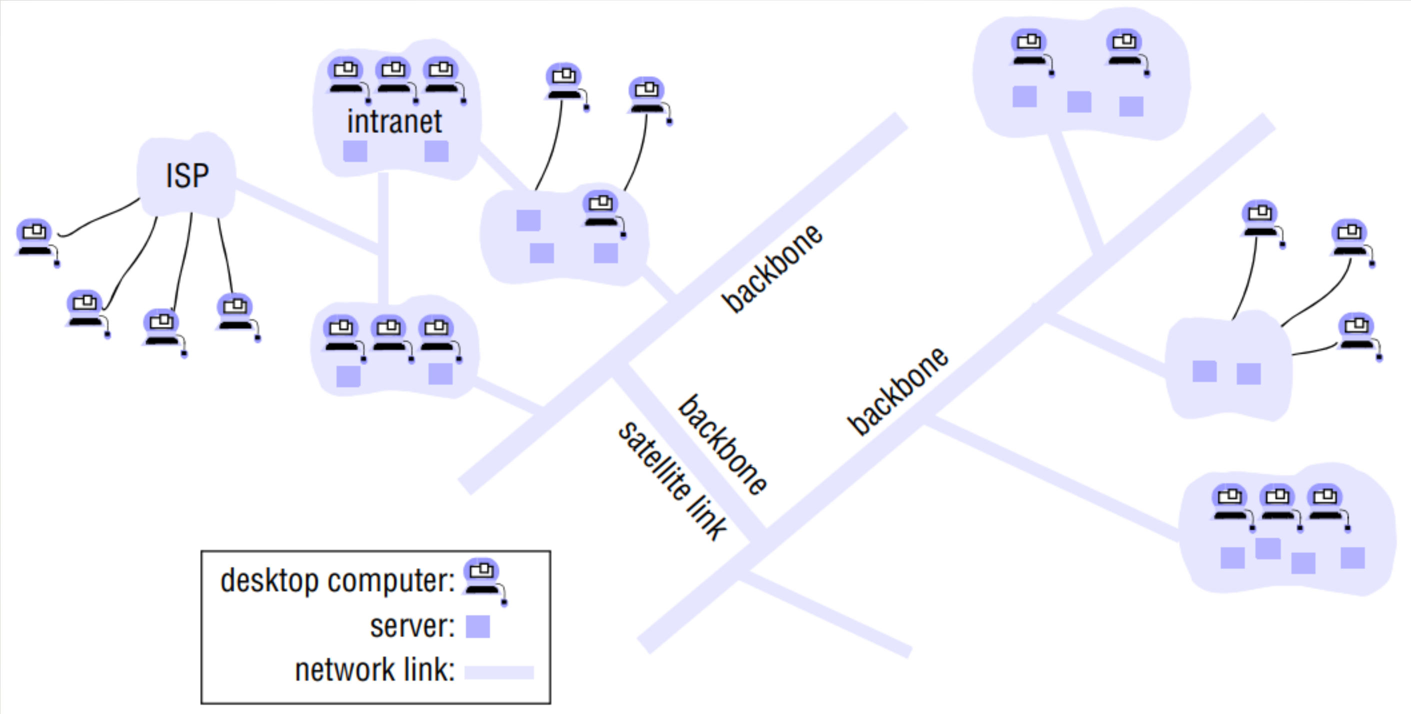
- **Incremental growth**: Computing power can be added in small increments. Modular expandability
- **Another deriving force**: the existence of large number of personal computers, the need for people to collaborate and share information.

1.2 Examples of Distributed Systems

- The Internet, Intranets, Enterprise Networks
- Mobile and Ubiquitous Computing
- Web services, Cloud computing, P2P systems, Sensor networks, Distributed databases

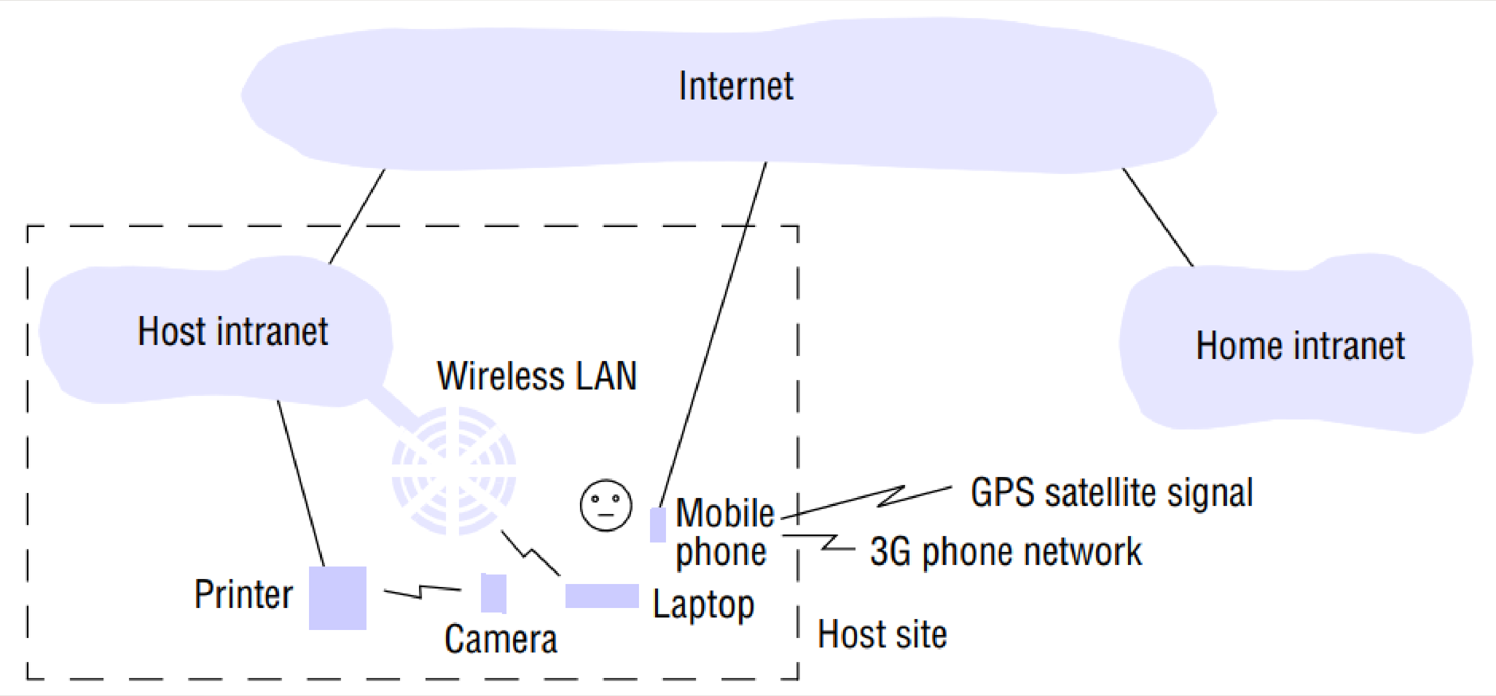
1.2 Examples of Distributed Systems

- A typical portion of the Internet



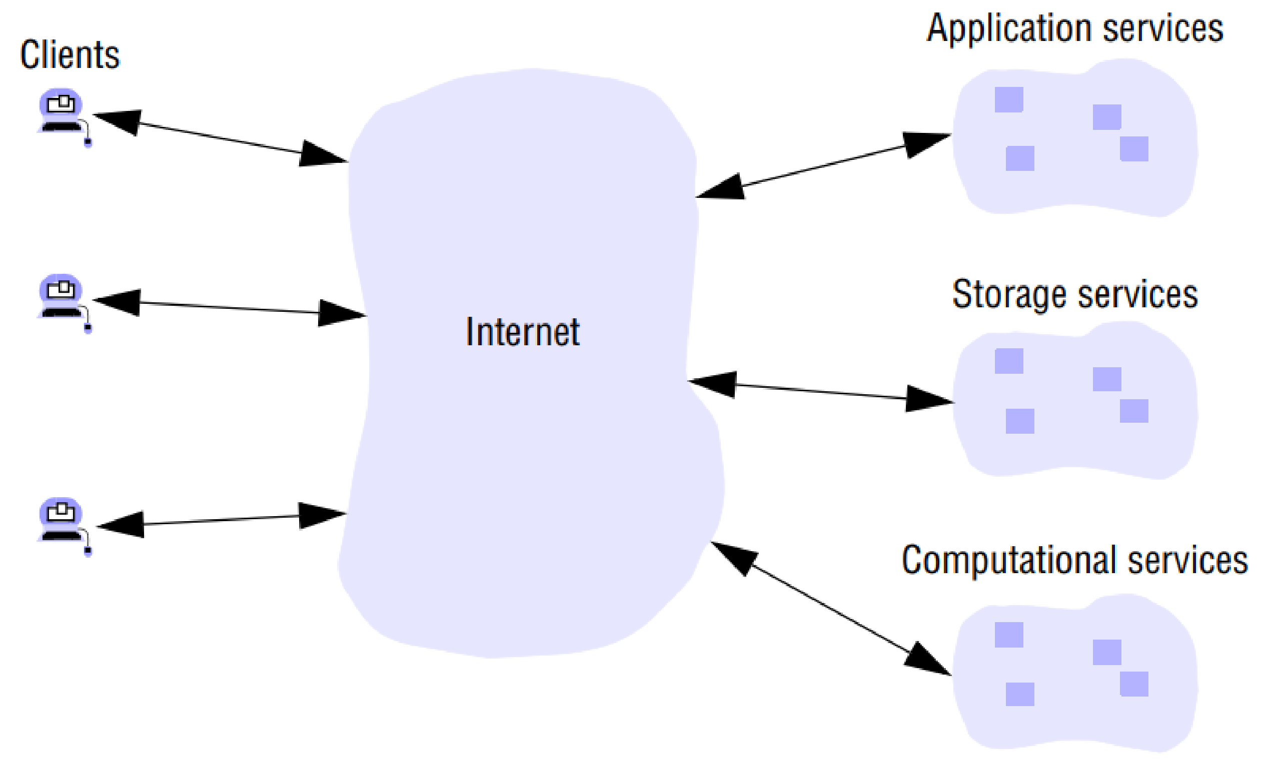
1.2 Examples of Distributed Systems

- Portable and handheld devices in a distributed system



1.2 Examples of Distributed Systems

- Cloud computing

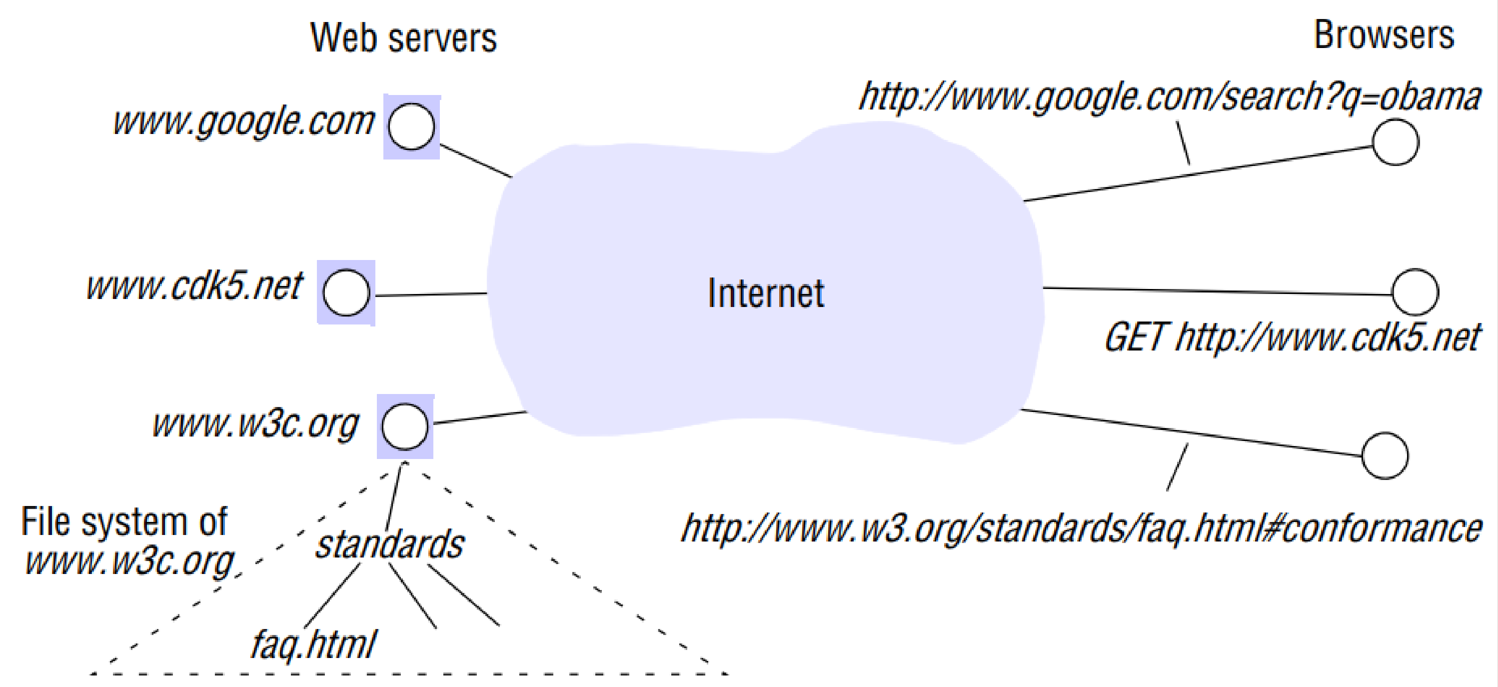


1.3 Resource Sharing and the Web

- Web as a platform for resource sharing
- Technologies: URLs, HTTP, Web browsers
- Programmatic access: Web services
- In general, HTTP URLs are of the following form:
  http:// servername [:port] [/pathName] [?query] [#fragment]

1.3 Resource Sharing and the Web

- Web servers and web browsers



1.4 Challenges

- Heterogeneity
- Openness
- Security enforcement
- Scalability
- Failure handling
- Concurrency management
- Transparency
- Quality of service

1.4 Challenges - Heterogeneity

- Heterogeneity (that is, variety and difference) applies to all of the following:
  - networks;
  - computer hardware;
  - operating systems;
  - programming languages;
  - implementations by different developers.

1.4 Challenges - Openness

- Definition of open systems
- Importance of standards and interoperability
- Portability of applications

1.4 Challenges - Security

- Confidentiality, Integrity, Availability
- Authentication and Authorization
- Challenges in open environments

1.4 Challenges - Scalability

- Controlling the cost of physical resources
- Controlling the performance loss
- Preventing software resources running out
- Avoiding performance bottlenecks

1.4 Challenges - Scalability

Growth of the Internet (computers and web servers)

Date	*Computers*	*Web servers*	*Percentage*
1993, July	1,776,000	130	0.008
1995, July	6,642,000	23,500	0.4
1997, July	19,540,000	1,203,096	6
1999, July	56,218,000	6,598,697	12
2001, July	125,888,197	31,299,592	25
2003, July	~200,000,000	42,298,371	21
2005, July	353,284,187	67,571,581	19

1.4 Challenges - Failure Handling

- Detecting failures
- Masking failures
- Tolerating failures
- Recovery from failures
- Redundancy
- Availability

1.4 Challenges - Concurrency

- Coordinating concurrent processes
- Need for synchronization and mutual exclusion

1.4 Challenges - Transparency

- *Access transparency*
- *Location transparency*
- *Concurrency transparency*
- *Replication transparency*
- *Failure transparency*
- *Mobility transparency*
- *Performance transparency*
- *Scaling transparency*

1.4 Challenges - Quality of service

- Reliability
- Security
- Performance
- Adaptability

REVIEW & Questions

- Recap of key concepts
- Questions